Dynamic modeling of hepatitis B virus infected hepatocytes using peripheral viral markers and validation with liver biopsy long-read DNA sequencing

Poster D-274

Cole Campton[1,2], David Pan[1], Mario Cortese[1], Jeffrey Wallin[1], Xiaobai Sun[2], Chi-Yi Chen[3], Chen-Hao Tseng[4,5], Scott Balsitis[1], Yao-Chun Hsu[4,5], Bryan Downie[1]

[1]Gilead Sciences, Foster City, CA USA; [2]Duke University, Computer Science, Durham, NC USA; [3]Division of Gastroenterology and Hepatology, Ditmanson Medical Foundation Chia-Yi Christian Hospital, Chia-Yi, Taiwan; [4]School of Medicine and Graduate Institute of Medicine, I-Shou University, Kaohsiung, Taiwan; [5]Division of Gastroenterology and Hepatology, E-Da Hospital, Kaohsiung, Taiwan;

Conclusions

- HBV DNA within hepatocytes exists as either covalently closed circular DNA (cccDNA) or as integrated DNA (intDNA).
- Functional Cure is defined as persistent undetectable levels of serum HBsAg.
- HBsAg is produced by both cccDNA and intDNA, but current methods to quantify intDNA require invasive liver biopsies.
- Non-linear mixed effects modeling was used to estimate infected cells using longitudinal peripheral viral biomarkers (HBV DNA, HBsAg, HBeAg, and HBcrAg).
- Predicted cccDNA and intDNA cell fractions from the model were validated against PacBio sequencing reads to demonstrate the utility of the model
- Model reflected HBV-host infection dynamics from serum measurements which correlated with liver biopsy-derived results.
- These results provide evidence for the utility of modeling of HBV infection dynamics in understanding cccDNA and intDNA disease burden through non-invasive means.

Plain Language Summary

A novel model was developed for chronic HBV using non-invasive serum biomarkers to capture the two forms of HBV DNA in infected liver cells. Matched liver biopsies were used to confirm the serum results. Experimental and modeled quantities were compared to establish the feasibility of this peripheral measurement/modeling approach as a proxy for invasive biopsies

References

1. Tu, T., Budzinska, M. A., Shackel, N. A., and Urban, S. (2017). Hbv dna integration: Molecular mechanisms and clinical implications. Viruses, 9(4).
2. Wooddell, C. I., Yuen, M. F., Chan, H. L., Gish, R. G., Locarnini, S. A., Chavez, D., Ferrari, C., Given, B. D., Hamilton, J., Kanner, S. B., Lai, C. L., Lau, J. Y. N., Schluep, T., Xu, Z., Lanford, R. E., and Lewis, D. L. (2017). Rnai-based treatment of chronically infected patients and chimpanzees reveals that integrated hepatitis b virus dna is a source of hbsag. Sci Transl Med, 9(409).
3. Zhao, X.-L., Yang, J.-R., Lin, S.-Z., Ma, H., Guo, F., Yang, R.-F., Zhang, H.-H., Han, J.-C., Wei, L., and Pan, X.-B. (2016). Serum viral duplex-linear dna proportion increases with the progression of liver disease in patients infected with hbv. Gut, 65(3):502–511.
4. Tu, T., Zehnder, B., Wettengel, J. M., Zhang, H., Coulter, S., Ho, V., Douglas, M. W., Protzer, U., George, J., and Urban, S. (2022). Mitosis of hepatitis b virus-infected cells in vitro results in uninfected daughter cells. JHEP Rep, 4(9):100514.
5. Tu T, Zhang H, Urban S. Hepatitis B Virus DNA Integration: In Vitro Models for Investigating Viral Pathogenesis and Persistence. Viruses. 2021; 13(2):180.
6. Pan, D., Soulette, C., Han, D., Ramirez, R., van Buuren, N., Lin, J., Tseng, C., Li, L., Fletcher, S., Mo, H., Feierbach, B., Suri, V., and Hsu, Y.-C. (2022). Decrease in the burden of integrated hepatitis b virus dna in chronic hepatitis b patients with minimally elevated alanine aminotransferase on tenofovir disoproxil fumarate as revealed by long-read dna sequencing. In European Association for the Study of the Liver (EASL), London, United Kingdom.

Acknowledgments: We extend our thanks to the patients, their families, and all participating investigators. This study was funded by Gilead Sciences, Inc. Editing and production assistance were provided by [company name], and funded by Gilead Sciences, Inc.

Disclosures: **CC, DP, MC, JW, SB, and DB** are employees of Gilead Sciences, Inc., and may own stock in Gilead Sciences, Inc.

Correspondence: Cole Campton, cole@colecampton.com

Introduction

Functional cure in chronic hepatitis B (CHB) is defined as persistent loss of detectable serum hepatitis B surface antigen (HBsAg) and HBV DNA. Integrated HBV DNA (intDNA) in hepatocytes is a major source of HBsAg, but current methods to quantify intDNA require invasive liver biopsies. We developed a mathematical model (**Figure 1**) to estimate the relative abundance of hepatocytes with intDNA or covalently closed circular DNA (cccDNA) using serum viral measurements.

Methods

The non-linear mixed-effects model was implemented in ICON Plc Nonlinear Mixed Effects Modeling software (NONMEM) and fit using stochastic approximate expectation maximum (SAEM) to data from CHB patients (**Table 1**) in a randomized, double-blind, placebo-controlled study of Tenofovir Disoproxil Fumarate (TDF) 300 mg/day or placebo over 3 years (NCT01522625).

Paired core liver biopsies (**Table 2**) collected per protocol at baseline and year 3 were used to determine observational relative cccDNA and intDNA content and for a novel comparison to modeled cccDNA and intDNA.

Table 1. Participant Counts by Treatment and Serostatus

Treatment	HBeAg-	HBeAg+
Placebo	57	19
TDF	62	11

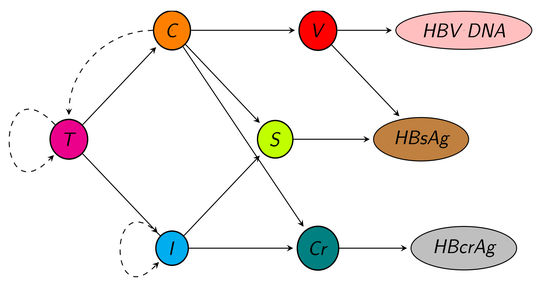
Table 2. Biopsy Long Read Sequencing Sub-study Participant Counts by Treatment and Serostatus

Treatment	HBeAg-	HBeAg+
Placebo	17	10
TDF	25	3

Model Overview

- Target hepatocytes (T) become infected by virions
- Infected cells are cccDNA (C) and intDNA (I) infected
- cccDNA and intDNA infected cells originate from uninfected cells at a rate p, interpretable as the double-stranded linear (dsl-)DNA versus relaxed-circular (rc-)DNA rate
- cccDNA is lost upon cell proliferation **[4]**
- intDNA is propagated into both daughter cells
- cccDNA produces virions (V), sub-viral particles (S) and Hepatitis core antigen (HBcAg, Cr)
- intDNA produces only sub-viral particles (S) and HBcAg (Cr)
- Virions are measured in serum as HBV DNA and HBsAg
- Sub-viral particles are measured in serum as HBsAg
- Hepatitis core antigen is measured in serum as Hepatitis core-related antigen (HBcrAg)

Figure 1. Diagram of HBV Infection Dynamics



Above, diagrammatic representation of the HBV model. Right, the ordinary differential equation (ODE) representation of the HBV model.
Each subscripted d parameter specifies death rates of related cells. r specifies the cellular proliferation rate. B is the infection rate constant. The value v_C is the daily per-capita rate of virion production by cccDNA infected cells. Parameters s and cr are similarly the daily rate of production for S and Cr by each cell. The shared clearance rate is c.

$$\dot{T} = r(T+2C)\left(1-\frac{T+C+I}{T_{max}}\right) - \beta VT - d_T T$$
$$\dot{C} = p\beta VT - rC\left(1-\frac{T+C+I}{T_{max}}\right) - d_C C$$
$$\dot{I} = (1-p)\beta VT + rI\left(1-\frac{T+C+I}{T_{max}}\right) - d_I I$$
$$\dot{V} = v_C C(1-EFF(t)) - cV$$
$$\dot{S} = s_C C + s_I I - cS$$
$$\dot{Cr} = cr_C C + cr_I I - cCr$$

Results

Fitted model results provided individual-specific estimates of viral protein production rates and infection cell quantities at each collection time. The quality of the model was assessed with three distinct methods:

1. Comparison of fitted model parameters (**Table 3**) to biological expectation and literature values
2. Assessment of biomarker fitting error visually (**Figure 2**)
3. Validation of model-predicted estimates for relative cccDNA and intDNA counts via comparison to long-read experimentally measured relative counts (**Figure 4**)

Fitted model parameters presented in (**Table 3**) relate to the indicated components of the dynamic model (**Figure 1**). Each parameter was fit to an individual subject level to either log-normal or logistic normal assumptions using maximum likelihood estimation (MLE). **Table 3** summarizes population level statistics for this study.

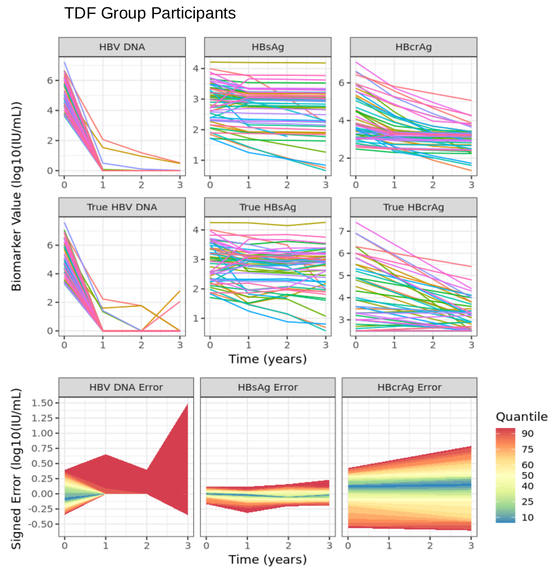
Table 3. Select Fitted Parameters

Parameter	Mean	Variance
Uninfected hepatic death rate (d_T)	$1.15 \cdot 10^{-6}$	$1.70 \cdot 10^{-21}$
cccDNA infected death rate (d_c)	$5.91 \cdot 10^{-3}$	$3.45 \cdot 10^{-3}$
intDNA infected death rate (d_I)	$7.76 \cdot 10^{-4}$	$1.13 \cdot 10^{-3}$
Rate of rcDNA verse dslDNA (p)	94.0%	0
Logistic proliferation rate (r)	$2.42 \cdot 10^{-2}$	$4.88 \cdot 10^{-17}$
cccDNA sub viral particle production rate (cr_C)	$5.79 \cdot 10^{-5}$	$9.53 \cdot 10^{-20}$
intDNA sub viral particle production rate (cr_I)	$3.59 \cdot 10^{2}$	$3.49 \cdot 10^{2}$
cccDNA HBcAg production rate (cr_C)	$1.23 \cdot 10^{-5}$	$7.63 \cdot 10^{-5}$
intDNA HBcAg production rate (cr_I)	$2.07 \cdot 10^{-8}$	$3.32 \cdot 10^{-24}$
cccDNA virion production rate (v_C)	$1.15 \cdot 10^{-5}$	$1.54 \cdot 10^{-4}$
TDF Efficacy (EFF)	100.%	$1.58 \cdot 10^{-4}$

Fitted model parameters (**Table 3**) reflect known biology and literature values. Viral protein rates indicate less HBcAg is secreted from intDNA per-capita, reflecting the expectation that dslDNA frequently linearizes at the core/pre-core ORF **[1]**.
Additionally, most sub-viral particles are produced by intDNA, in-line with biological expectations for this HBeAg- population **[2]**.

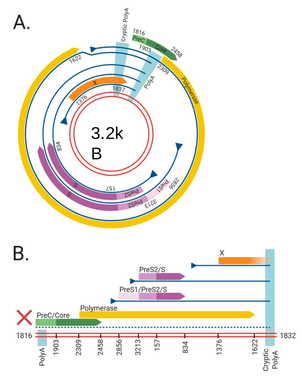
The fitted parameters suggest 6% of virions contain dslDNA, close to the 7% value reported elsewhere**[3]**.

Figure 2. Participant Level Model Fit



The figure above displays longitudinal biomarkers column-wise for each of the serum measured values: HBV DNA, HBsAg and HBcrAg. The first row displays model fitted values while the second row displays the true measured values, where identical participants have been assigned identical colors. Some changes are not captured by the model fit, namely because participant level parameters are regularized by log-normality assumptions to prevent over-fitting.
The errors produced by modeling are displayed on the third row, summarized by the residual error at each quantile.

Figure 3. Hepatitis B Genome



The two forms of viral DNA, adapted from **[5].** In both figures blue lines represent viral mRNAs and triangles indicate transcriptional promoters. Colored arrows indicate open reading frames.
A: The HBV genome of replication viable cccDNA has four overlapping open reading frames encoding for seven viral proteins
1. HBcAg: Hepatitis core antigen
2. HBeAg: Hepatitis E antigen: Inhibits apoptosis
3. HB pol: HBV polymerase: Initiates replication
4. HBx: Hepatitis x protein: Enhances transcription

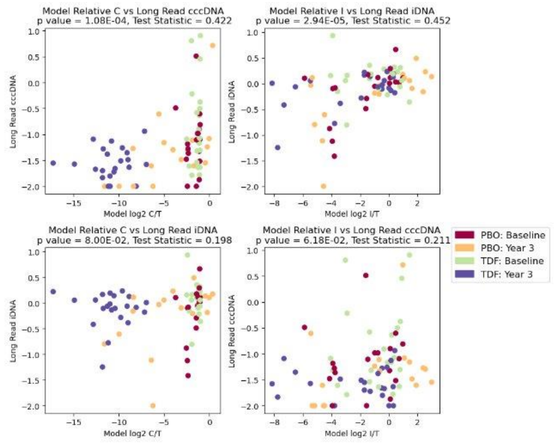
5-7. HBsAg: Hepatitis surface antigen, three varieties

B. dslDNA frequently linearizes at the core/pre-core ORF, leading to absence of the PreC/Core promoter and therefore less HBcAg from intDNA.

Liver biopsies from individuals were sequenced with long read techniques using a novel quantitative method for identifying Viral Integrations and Translocations (VALIANT) **[6]**. In short, the length of DNA collected from biopsies exceeds the 3.2 kB length of the HBV genome presenting in (**Figure 3.A**), making it possible to distinguish chimeric DNA reads of intDNA integration events from cccDNA reads. By normalizing to genes with known frequencies within human liver cells it is possible to experimentally measure cccDNA and intDNA relative cells counts within each biopsy.

To validate predicted cell populations, we computed correlation between experimental long-read derived relative cell counts with the equivalent model computed quantities at baseline and year 3. The results of this comparison are presented in (**Figure 4**). Results are inherently limited by noise contributed by low long-read sequencing depth and the variability of biopsy collection site.

Figure 4. Long-Read Sequencing verse Model Computed Relative Infection



Correlations between experimentally long-read derived relative cell counts and model derived cell counts are displayed above, colored by treatment group and time point. In the top row, as expected there is a statistically significant correlation between experimental and model derived counts of the same cell types. The bottom rows demonstrates that there is no statistically significant correlation between experimental and model derived counts for opposite cell types, as expected.

Conclusions

There are significant correlations between modeled and observed cccDNA cell population (Pearson's r=.422, p<.001) as well as intDNA cell population (Pearson's r=.452, p<.001).
The correlation between modeled and experimental quantities indicates that the longitudinal model is able to distinguish between the disease burden of study participants.
Model results reflect biological expectation and can reproduce liver biopsy-derived results from only serum measurement. These results provide evidence for the utility of modeling of HBV infection dynamics in understanding cccDNA and intDNA disease burden through non-invasive means for HBeAg- CHB patients.